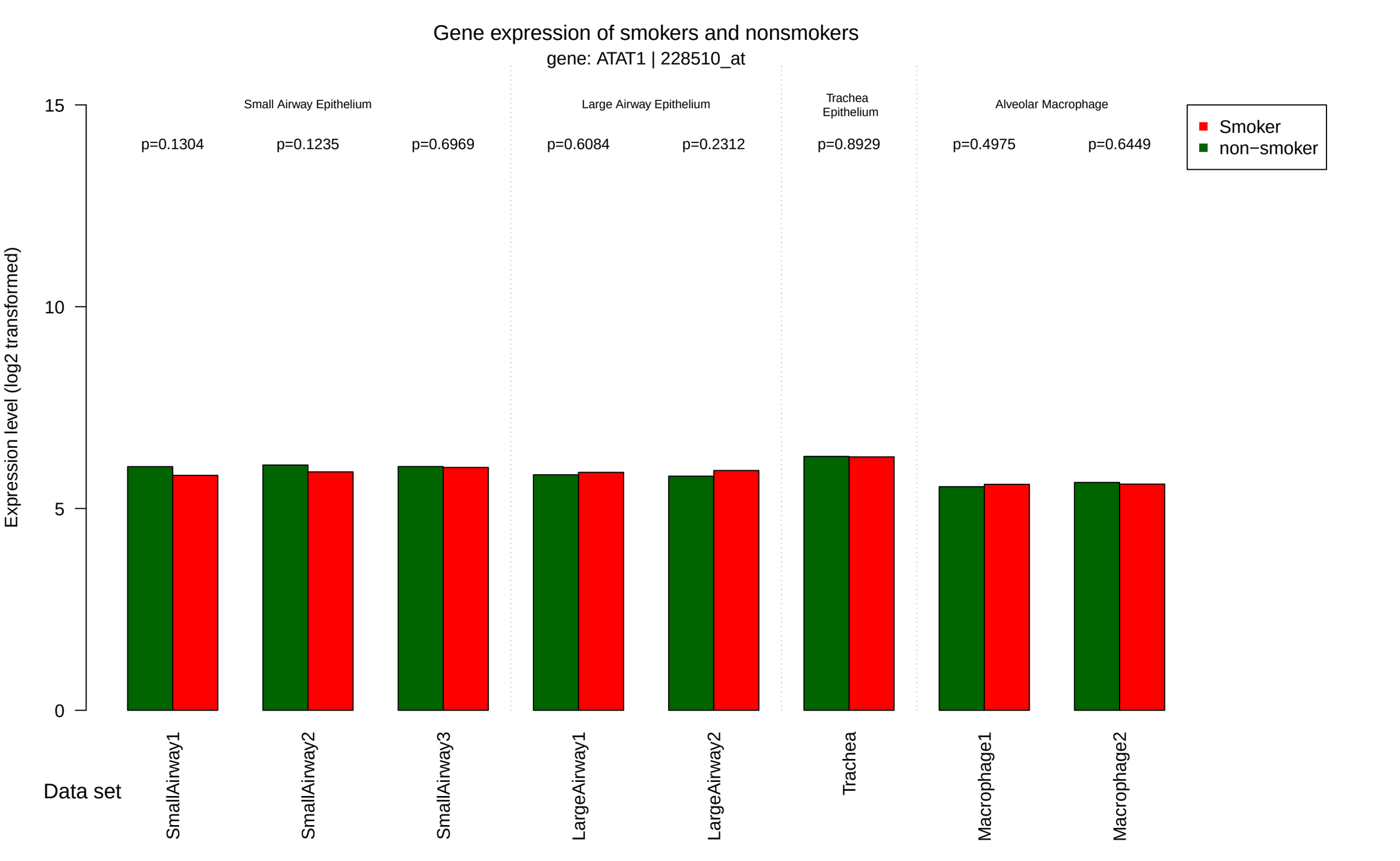
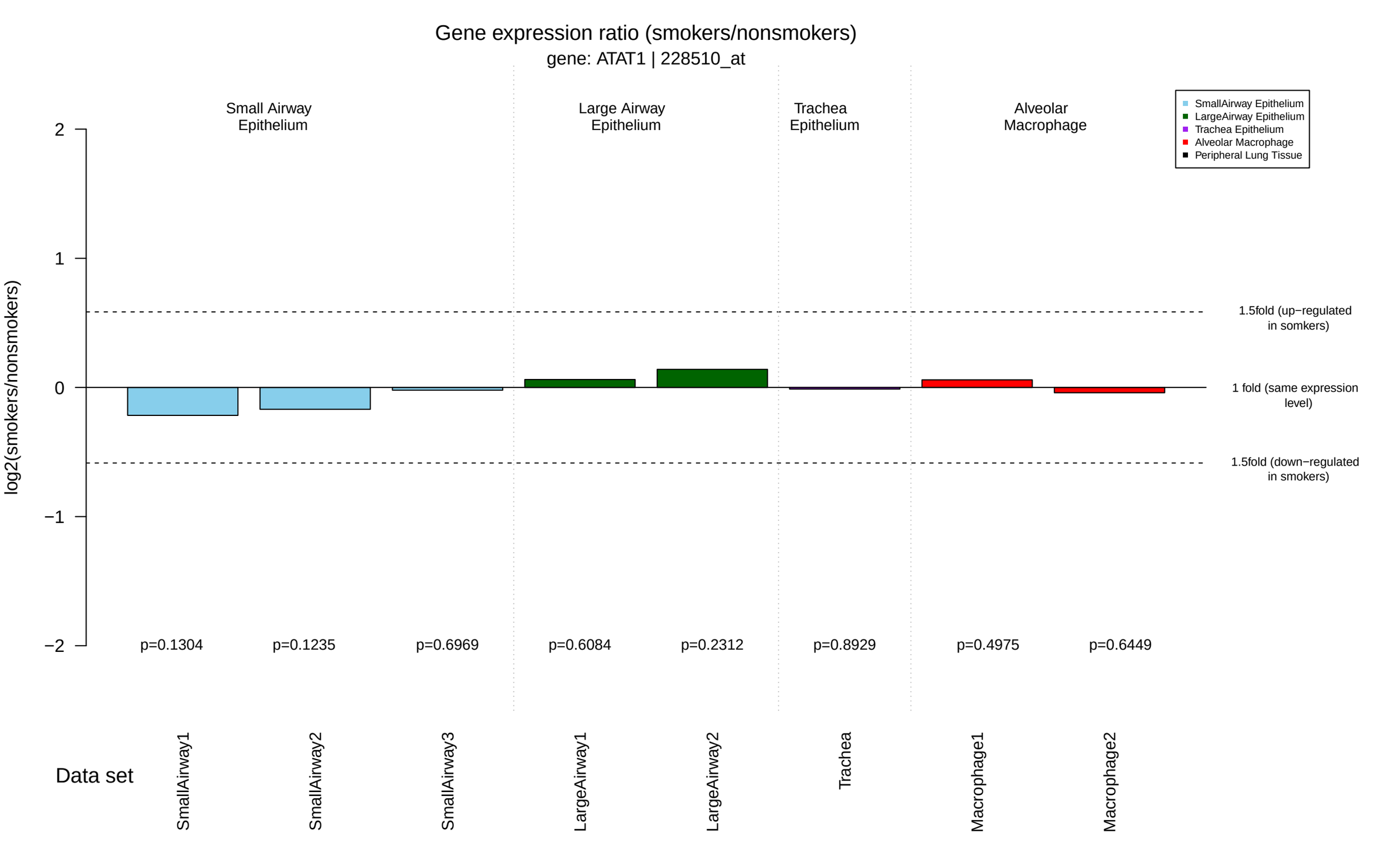
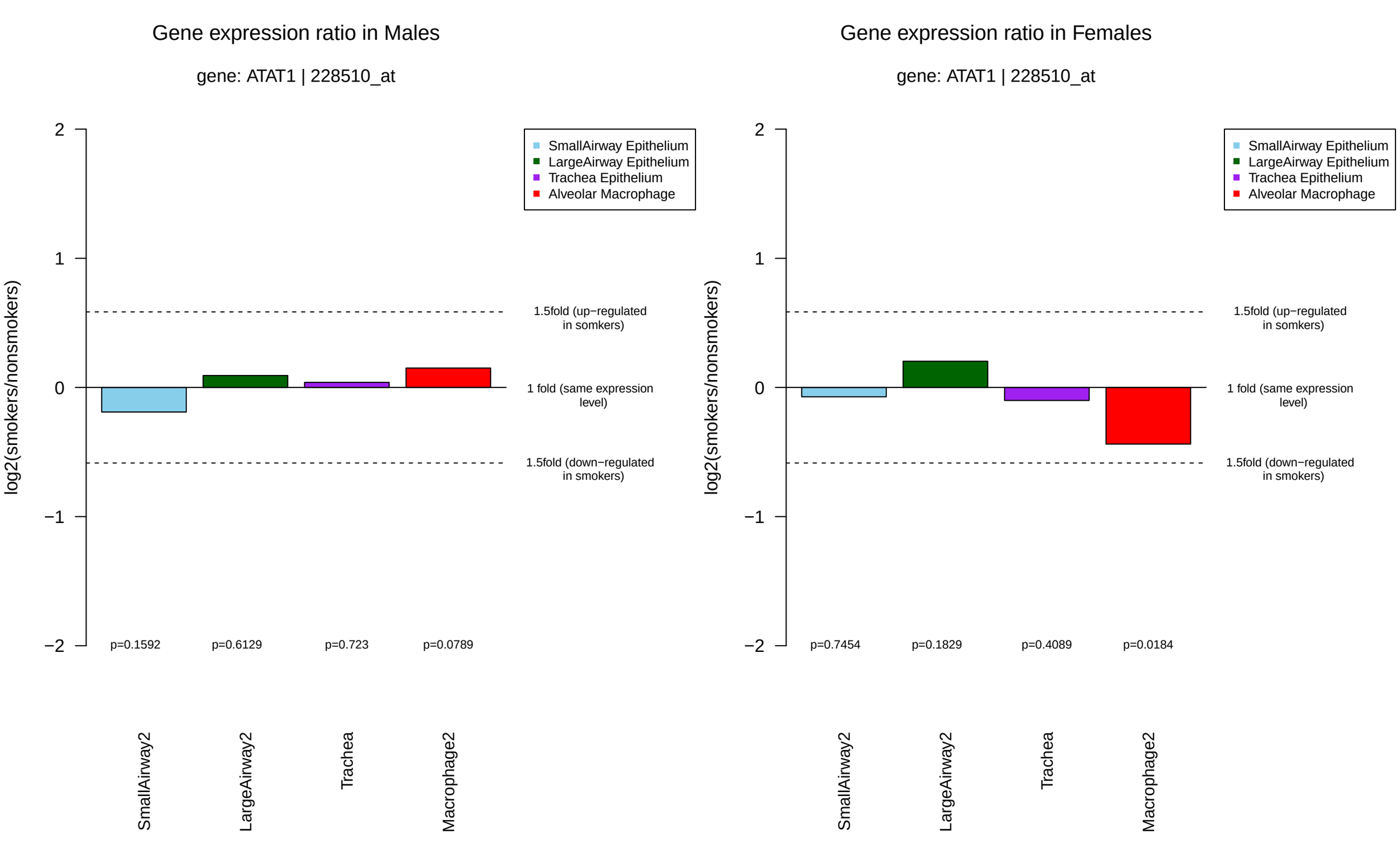
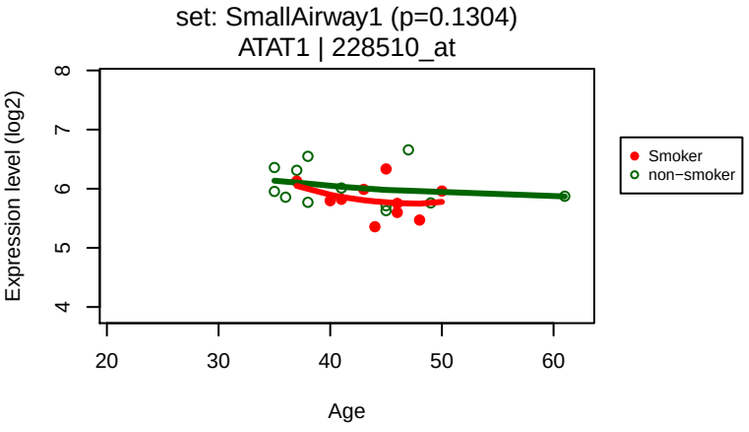
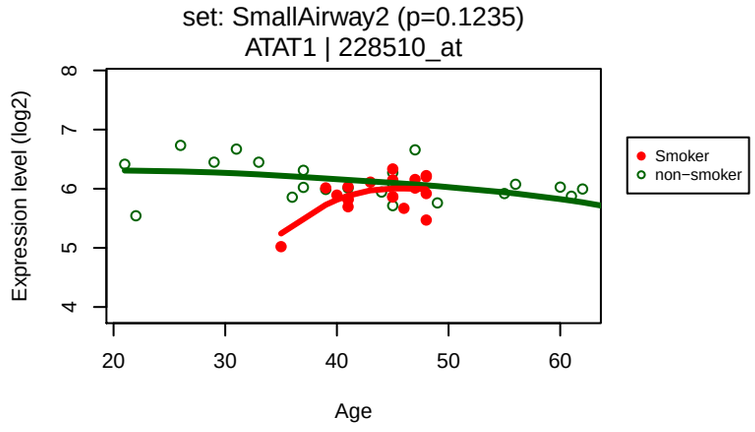
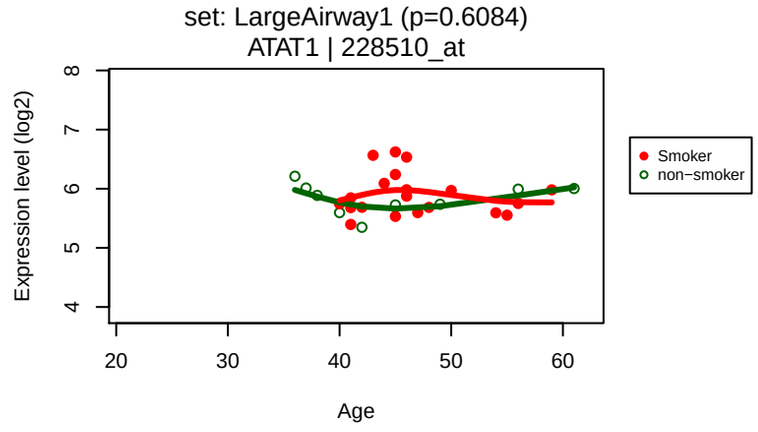

Gene expression of smokers and nonsmokers
gene: ATAT1 | 228510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1304
p=0.1235
p=0.6969
p=0.6084
p=0.2312
p=0.8929
p=0.4975
p=0.6449
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

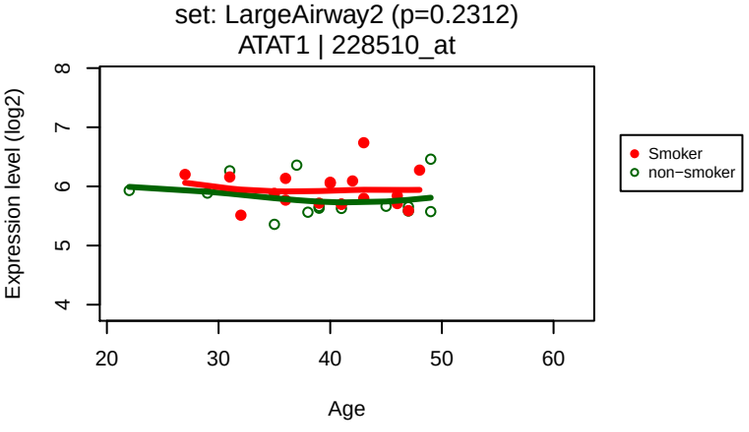

Gene expression ratio (smokers/nonsmokers)
gene: ATAT1 | 228510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1304
p=0.1235
p=0.6969
p=0.6084
p=0.2312
p=0.8929
p=0.4975
p=0.6449
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

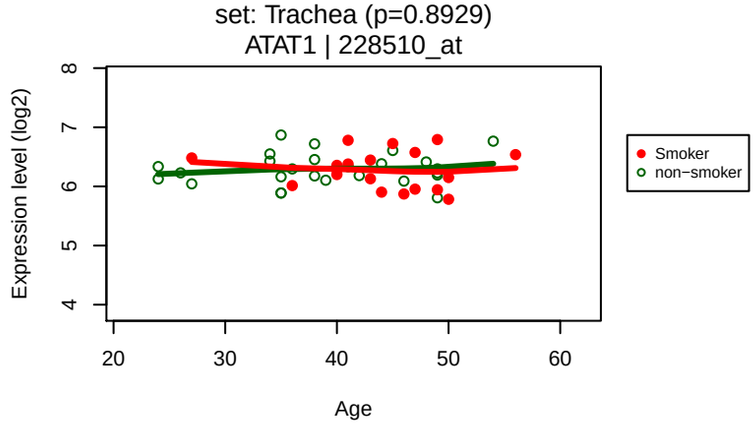

Gene expression ratio in Males
gene: ATAT1 | 228510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1592
p=0.6129
p=0.723
p=0.0789
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATAT1 | 228510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7454
p=0.1829
p=0.4089
p=0.0184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

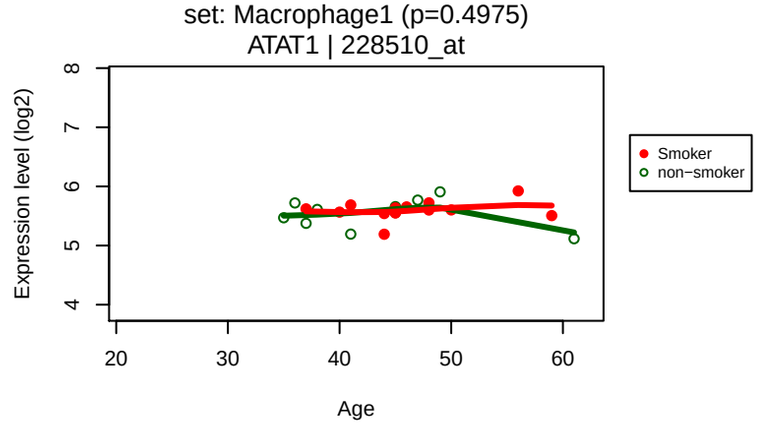

set: SmallAirway1 (p=0.1304)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker

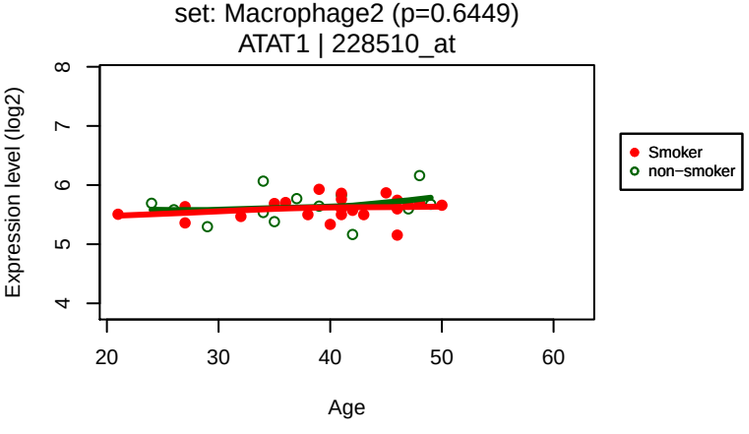

set: SmallAirway2 (p=0.1235)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker

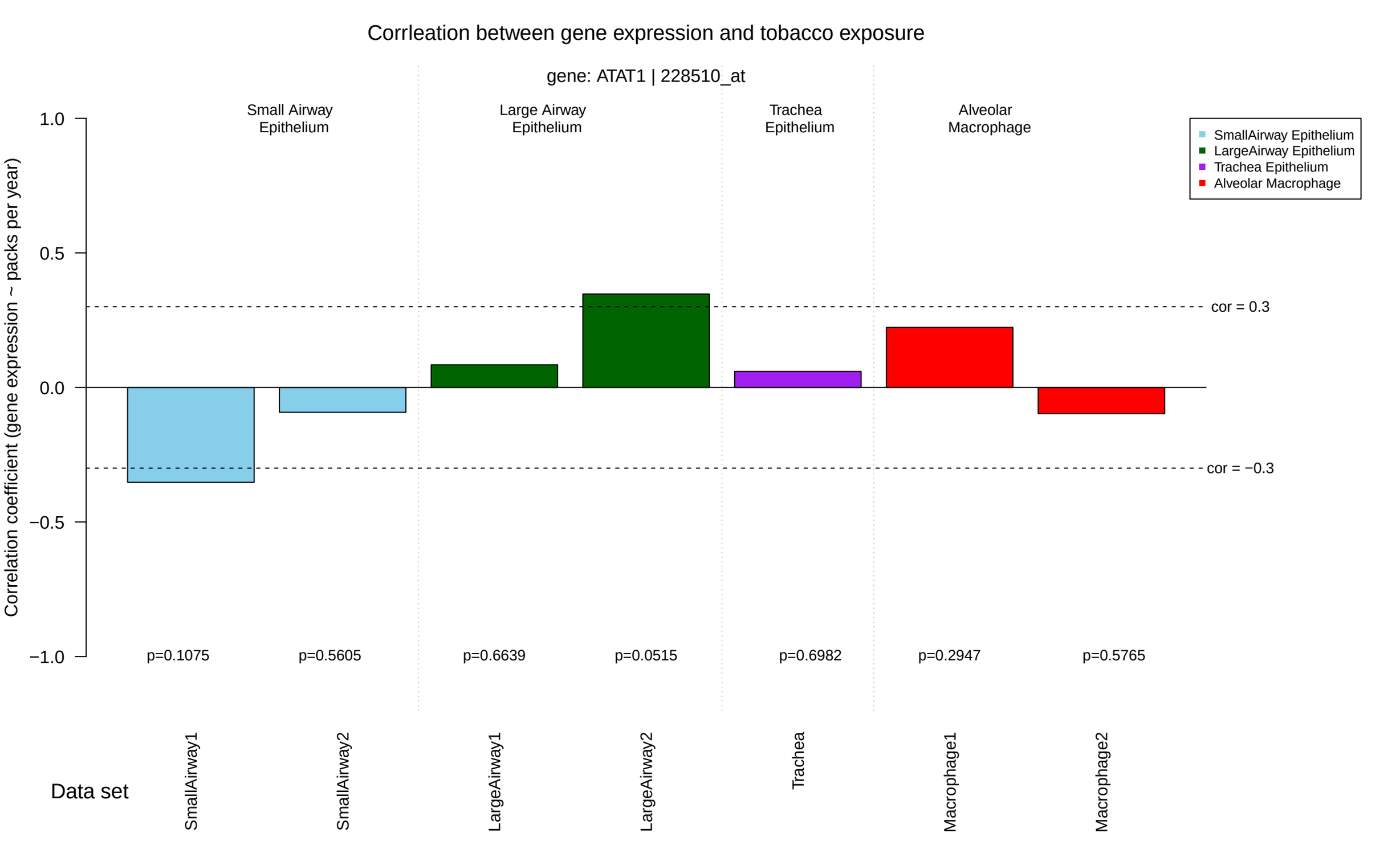

set: LargeAirway1 (p=0.6084)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2312)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8929)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4975)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6449)
ATAT1 | 228510_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATAT1 | 228510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1075
p=0.5605
p=0.6639
p=0.0515
p=0.6982
p=0.2947
p=0.5765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2